

Phenological Classification of Soybeans: Comparison between Convolutional Neural Networks and Random Forest with Integration into a Mobile Application

Abstract—Soybean desiccation is an essential practice for maximizing productivity, with the R7 phenological stage, according to the Fehr and Caviness system, being the ideal time for its execution. Traditional identification methods based on manual observation are subjective and prone to errors, potentially compromising seed quality and crop yield. Studies show that desiccation at the R7.1 stage significantly improves seed quality, while premature applications, such as at R6, impair performance. With the advancement of artificial intelligence (AI), automated methods such as Convolutional Neural Networks (CNNs) and the Random Forest algorithm emerge as effective solutions for the phenological classification of soybeans. CNNs extract features from agricultural images, utilizing architectures such as ResNet50, InceptionV3, and VGG16, while Random Forest analyzes images and agronomic data, standing out for its robustness and generalization capability. The integration of these technologies with mobile applications developed in frameworks like Flutter facilitates producers' access to practical and efficient tools, promoting precision agriculture. The use of well-labeled datasets is fundamental for training the models, ensuring high accuracy in the identification of phenological stages beyond R7. Thus, automating this process reduces human errors, improves seed quality, and boosts sustainability and efficiency in soybean management.

Keywords—Soybean Desiccation; Phenological Classification; Convolutional Neural Networks; Random Forest; Mobile Applications; Precision Agriculture

I. INTRODUCTION

Desiccation of soybean is a common practice aimed at maximizing both yield and harvest quality. Determining the optimal timing for this application is based on identifying phenological stages, which indicate the plant's maturity level. The traditional method, conducted through manual observation, relies heavily on the experience of the farmer or the responsible technician. However, this subjective approach can lead to errors that compromise decision-making. The R7 phenological stage is widely recognized as the most appropriate time for desiccation, according to the coding system established by Fehr and Caviness. The phenological classifications described in this system include subdivisions that exhibit specific characteristics,

which are essential for the application of Artificial Intelligence (AI).

TABELA I
SUMMARY OF SOYBEAN GROWTH STAGES

Stage	Name	Description
R1	Beginning Bloom	One open flower at any node on the main stem.
R2	Full Bloom	One open flower at one of the two uppermost nodes on the main stem with a fully developed leaf.
R3	Beginning Pod	Pod 5 mm long at one of the four uppermost nodes on the main stem with a fully developed leaf.
R4	Full Pod	Pod 3 cm long at one of the four uppermost nodes on the main stem with a fully developed leaf.
R5	Beginning Seed	Seed 3 mm long in a pod at one of the four uppermost nodes on the main stem with a fully developed leaf.
R6	Full Seed	Pod containing green seeds that fill the pod cavity at one of the uppermost nodes on the main stem with a fully developed leaf.
R7	Beginning Maturity	One normal pod on the main stem that has reached its mature pod color
R8	Full Maturity	95% of the pods have reached their mature pod color

As demonstrated in [4], the application of herbicides at the R7.1 stage brings significant benefits to seed quality, positively impacting germination and vigor when compared to the R7.3 stage. These results highlight the importance of accurately identifying the ideal timing for desiccation in order to maximize seed performance and contribute to the final productivity and quality of soybeans. According to [12], the application of desiccants at different developmental stages of soybean affects productivity and physiological seed quality, with significantly lower values observed when desiccants are applied at the R6 stage, demonstrating a negative impact on yield. On the other

hand, [13] reported that desiccant application at the R7.1 stage allowed for harvest anticipation of up to six days while maintaining high germination rates (90% and 92%) compared to the control (76%). Furthermore, desiccation did not compromise seed productivity, positioning itself as an efficient management alternative in seed production fields.

As discussed in [3], the timing of desiccant application directly influences post-harvest attributes of soybean grains. Anticipating the process in relation to physiological maturity results in a lower thousand-grain weight and higher bulk density. Additionally, early applications result in reduced oil content. However, desiccation performed at the recommended stage contributes to minimizing damage caused by pathogens, reinforcing the importance of adhering to the appropriate phenological stages.

Artificial Intelligence (AI) plays an increasingly important role in the advancement of precision agriculture, and computer vision has enabled the automation of complex tasks such as phenological classification, which can be significantly enhanced by these technologies. Convolutional Neural Networks (CNNs) have shown favorable performance in image classification tasks, making them an efficient solution for such applications. The integration of AI into mobile applications with user-friendly interfaces increases accessibility, facilitating adoption by farmers, reducing operational costs, and fostering sustainable development in agriculture. This technology can be incorporated into production chain management systems, contributing across various operations. However, as pointed out in [2], understanding the specific characteristics of each AI approach is fundamental to ensure its proper application.

Artificial Neural Networks (ANNs) have been widely applied and successfully implemented across different sectors. Among their most relevant applications is image recognition and classification, allowing for the identification and categorization of visual content into specific classes [8]. When comparing models, it is important to note that despite the high accuracy of CNNs, their complexity and large data requirements may pose limitations. Conversely, the Random Forest (RF) algorithm, a decision tree-based learning model, has been successfully applied to image classification tasks. In [14], a CNN model using `imageClassificationMulti` was shown to effectively classify soybean plants with high accuracy, although it presented some limitations due to visually similar crops. In another study [15], CNNs were used to detect soybean diseases, but challenges were encountered in distinguishing between visually similar disease symptoms, which may lead to misclassification. These issues highlight the importance of a robust and high-quality dataset, with well-labeled images that cover all possible variations. Nevertheless, CNNs can be effectively trained when

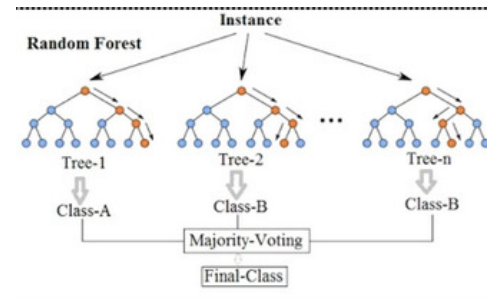


Fig. 1. The Random Forest (RF) algorithm

the dataset is diverse and capable of capturing relevant visual nuances.

This study proposes a comparison between three different CNN architectures and the Random Forest model to evaluate their performance in phenological classification tasks. The goal is to identify the most effective approach for accurately classifying phenological stages, considering model accuracy, ease of implementation, and data requirements. The CNN models used—ResNet50, InceptionV3, and VGG16—were chosen due to their proven efficiency in various computer vision contexts. The Random Forest (RF) model was selected for its strong performance in classification tasks, interpretability, and lower susceptibility to overfitting. This comparison enables a comprehensive analysis to determine the most suitable tool for phenological stage classification in soybean crops.

TABELA II
UMA TABELA EXEMPLO

One	Two
Three	Four

II. CONCLUSÃO

Descreva aqui as conclusões sobre o trabalho apresentado.

AGRADECIMENTOS

Coloque aqui seus agradecimentos.

REFERÊNCIAS